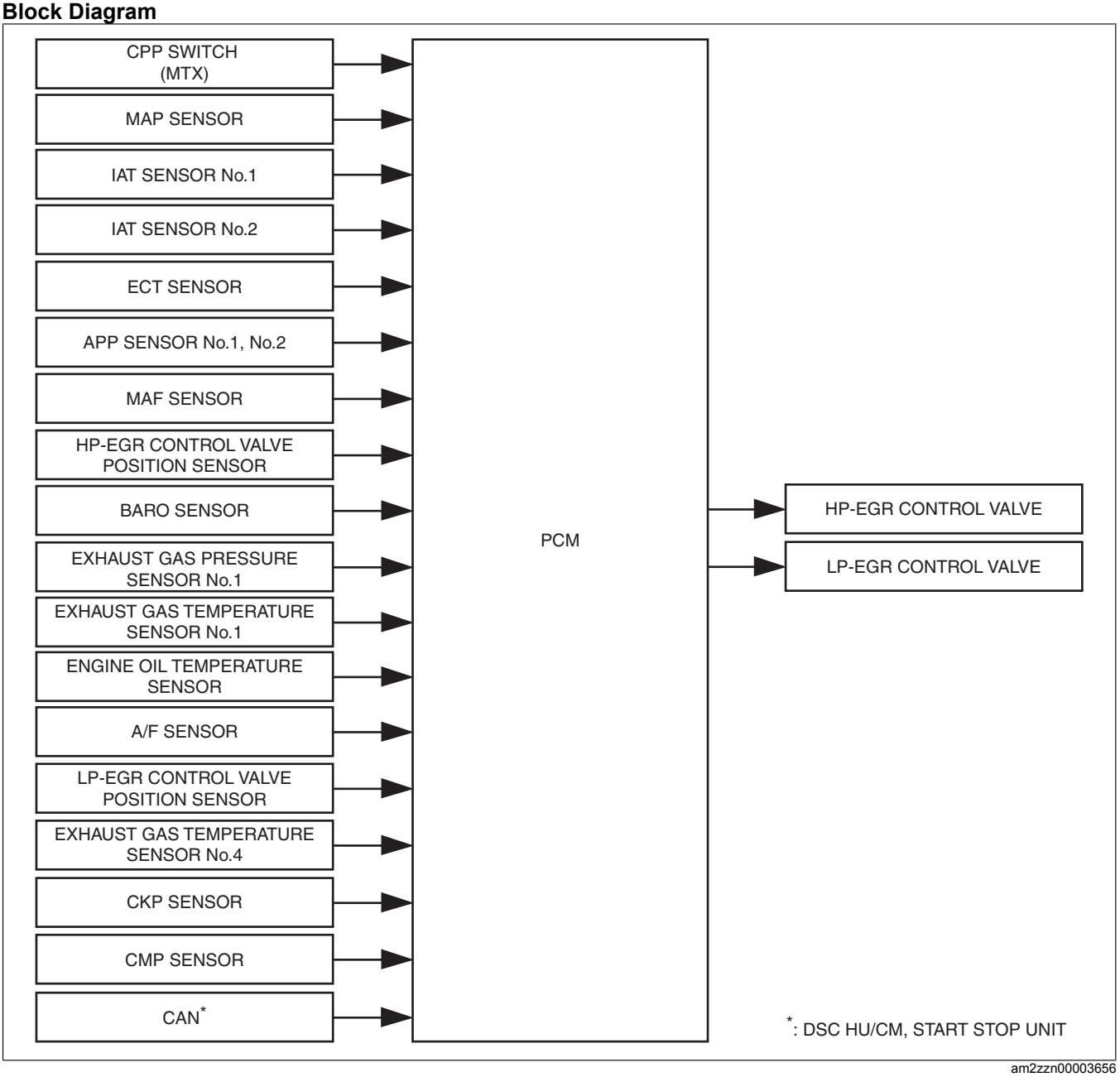
EGR CONTROL [SKYACTIV-D 1.5]

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Outline

- To recirculate an optimum amount of exhaust gas to the combustion chamber according to engine operation conditions, the PCM performs duty control of the HP-EGR control valve and LP-EGR control valve to open/closed the valve.
- During EGR control, the intake air temperature increases by the recirculated exhaust gas which affects the combustion temperature. To maintain an appropriate intake air temperature, the opening angle of the HP-EGR control valve and LP-EGR control valve is controlled according to the driving environment/engine conditions.
- During fast idle increase and DPF regeneration control, the HP-EGR control valve is controlled so that it does not operate to prevent the post-injected fuel from circulating to the intake side.

Block Diagram



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Operation

- Based on the input signal, the PCM calculates the required intake air amount and the EGR amount to achieve the oxygen concentration.
- The HP-EGR control valve and LP-EGR control valve is duty-controlled to open/close so that the EGR amount follows the targeted amount.

  Target oxygen concentration

  — The target oxygen concentration is calculated by adding corrections to the basic target oxygen concentration which is calculated based on the fuel injection amount.

Target EGR amount

— The target EGR amount is calculated by target intake air oxygen concentration, exhaust gas oxygen concentration, and charging efficiency.

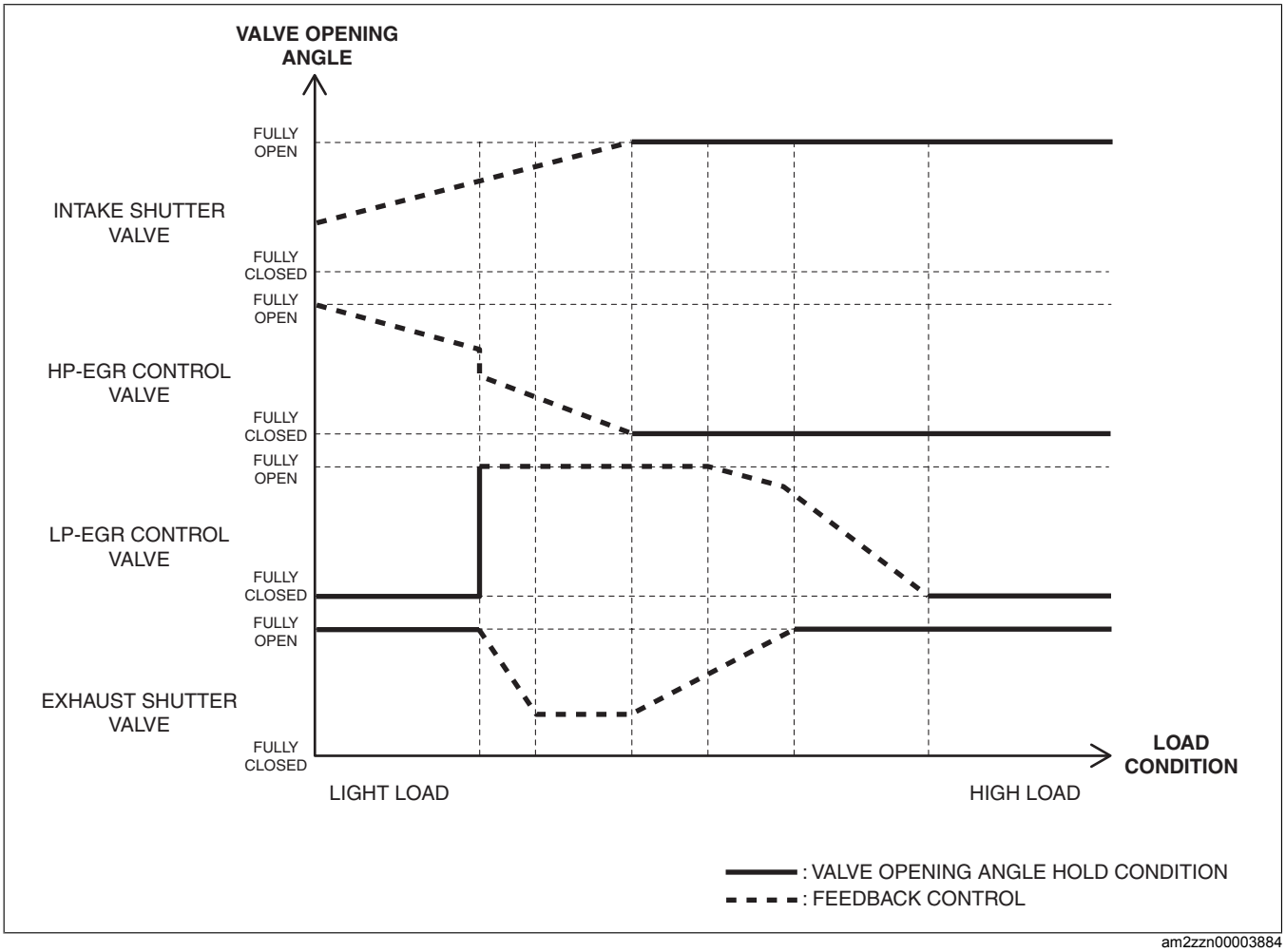
Actual EGR amount

— Calculated by the MAF sensor, HP-EGR control valve, and the charging efficiency.

Valve open/close operation overview during vehicle travel

- The PCM controls each valve during vehicle travel, while at the specified ambient temperature, complete engine warm-up, and stable driving conditions, as follows.*

* : The valve open/close condition varies depending on the engine operation conditions.



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- If the ambient and engine coolant temperatures are low, an increase correction in the EGR amount of the HP-EGR system (HP-EGR control valve and intake shutter valve) is made, and a decrease correction in the EGR amount of the LP-EGR system (LP-EGR control valve and exhaust shutter valve) is made to increase the intake manifold intake air temperature (intake air temperature sensor No.2).

Operation during DPF regeneration

- The PCM controls the LP-EGR control valve and intake shutter valve duty ratios according to the DPF regeneration conditions.
  - — Intake shutter valve: Duty control
  - — HP-EGR control valve: Maintained at fully closed
  - — LP-EGR control valve: Duty control*1
  - — Exhaust shutter valve: Maintained at fully open

*1 : Because the exhaust gas temperature is high during compulsory DPF regeneration, the LP-EGR control valve is fully closed to prevent combustion difficulties caused by the exhaust gas being recirculated.